

1. Project Overview

Project Title: Monitoring a Containerized URL Shortener Webservice

Objective: To build, containerize, and monitor a URL shortener service using **Docker**, **Prometheus**, and **Grafana** for performance tracking, visualization, and alerting.

2. Key Stakeholders

<i>Stakeholder</i>	<i>Role / Responsibility</i>	<i>Interest / Expectations</i>	<i>Influence / Power</i>	<i>Engagement Strategy</i>
<i>Project Developer</i>	Developer, DevOps Engineer	To successfully build, containerize, and monitor the URL shortener webservice. Wants to gain hands-on DevOps experience.	High	Responsible for full implementation, integration, and documentation.
<i>Instructor</i>	Evaluator and guide	Expects a well-structured, functional project demonstrating understanding of containerization, monitoring, and alerting.	High	Regular progress updates, clear documentation, demonstration of working stack.
<i>End Users</i>	Use and test the webservice	Expect the webservice to shorten URLs efficiently and show accurate metrics in dashboards.	Medium	Provide a simple, user-friendly API and clear test instructions.

<i>DevOps Tools (Prometheus & Grafana)</i>	Monitoring & visualization tools	Must be configured correctly to collect, visualize, and alert on metrics.	Medium	Continuous integration and testing to ensure proper metrics exposure and visualization.
<i>Database (SQLite)</i>	Data storage for URL mappings	Must persist data correctly even after container restarts.	Medium	Ensure volume persistence and data integrity through Docker volumes.
<i>Docker Environment</i>	Containerization platform	Needs to orchestrate all services (App, Prometheus, Grafana) correctly.	High	Maintain clear and modular Dockerfile and docker-compose.yml configurations.
<i>Network / System Resources</i>	Infrastructure dependency	Should support container communication, metric scraping, and dashboard access.	Medium	Allocate sufficient local resources; monitor CPU/memory usage.

3. Stakeholder Interest vs. Influence Matrix

<i>Category</i>	<i>Stakeholders</i>	<i>Management Approach</i>
<i>High Power, High Interest</i>	Developer, Instructor	Manage closely — provide regular progress demos, maintain quality and documentation.

<i>High Power, Low Interest</i>	Docker Platform	Keep satisfied — ensure system stability and performance.
<i>Low Power, High Interest</i>	End Users, Testers	Keep informed — provide access to the webservice and visual dashboards.
<i>Low Power, Low Interest</i>	External Viewers / Observers	Monitor only — share summary results in the final report.

4. Summary

This stakeholder analysis ensures that all participants and components in the DevOps pipeline from developer to monitoring tools are aligned. The main focus is maintaining clear communication, technical stability, and demonstrable monitoring insights through Prometheus and Grafana dashboards.